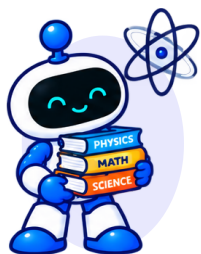


Should we deploy? And how much data do we need?

Applied hypothesis testing and power analysis for AI engineers

Who I am

Bruno Lannoo: ML Engineer @ JetBrains



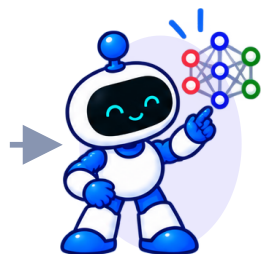
Physics student

loves making maths intuitive



Software developer

knows which details help the product

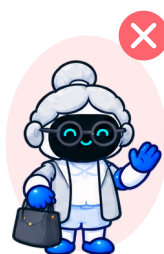


ML

engineer

needs the stats for evals

Who I hope you are!



Not

my grandma

does not know AI



Not

5 year olds

does not multiply



Some

AI builders

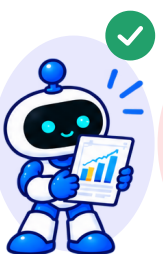
wide awake



Mostly

engineers

right at home



Some

stats savvy

should be a breeze



Not

statisticians

too low rigour



A **technical** presentation,
designed to be **accessible**.

If this reads as slop to you:
Sorry. I like it. And somehow it
made these slides more work.



but more fun



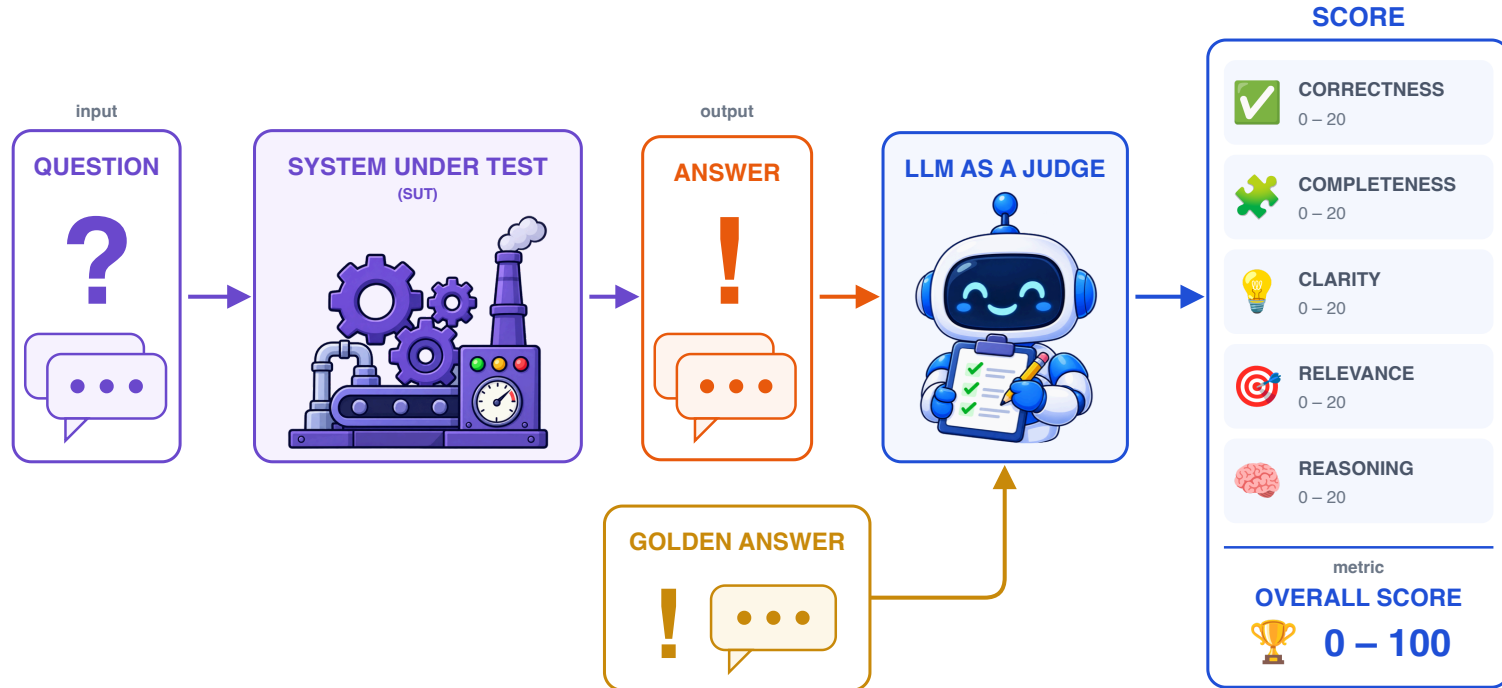
The two questions

1. Should we deploy? `hypothesis testing`
2. How much data do we need? `power analysis`

The sections of this talk

- ① The problem ② Admitting noise ③ Hypothesis testing ④ Power analysis

Scoring one answer



Today it's a judge score, but the story fits **any metric of your system**:



latency



cost



SWE-bench % resolved



thumbs-up rate



In many projects I have seen this rule:

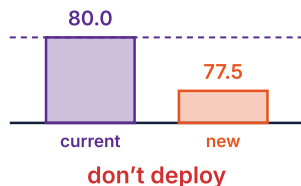
SUT A (current in prod)

> or < ?

SUT B (new candidate)

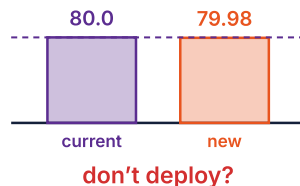
Scenario B1

clearly behind



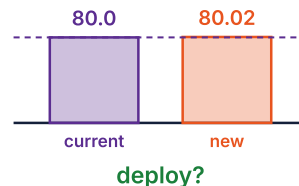
Scenario B2

barely behind



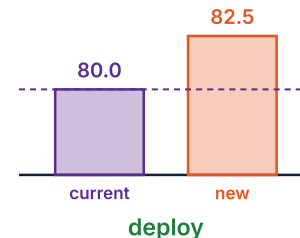
Scenario B3

barely ahead



Scenario B4

clearly ahead

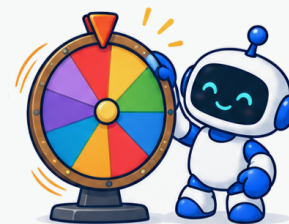


with maybe a gut-feel exception for too-small differences

Flip-flopping: re-run the same eval, and the winner swaps.

	SUT A (current)	SUT B (new)	
run 1	80.00	80.02	✓ ship B
re-run	80.01	79.98	✗ ship A back?

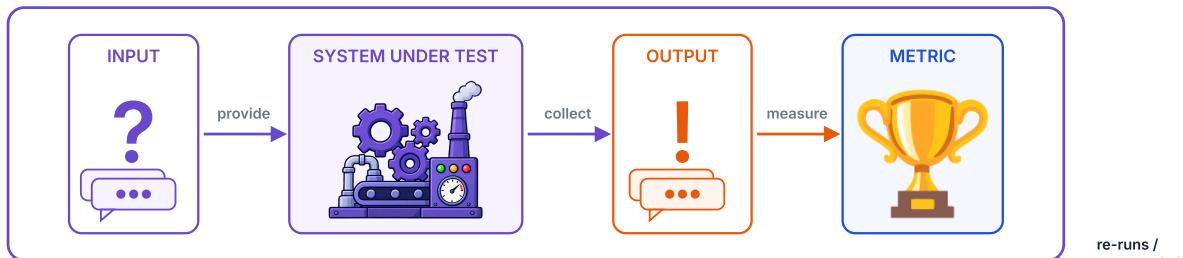
Nothing changed between the runs — the system just is not deterministic.



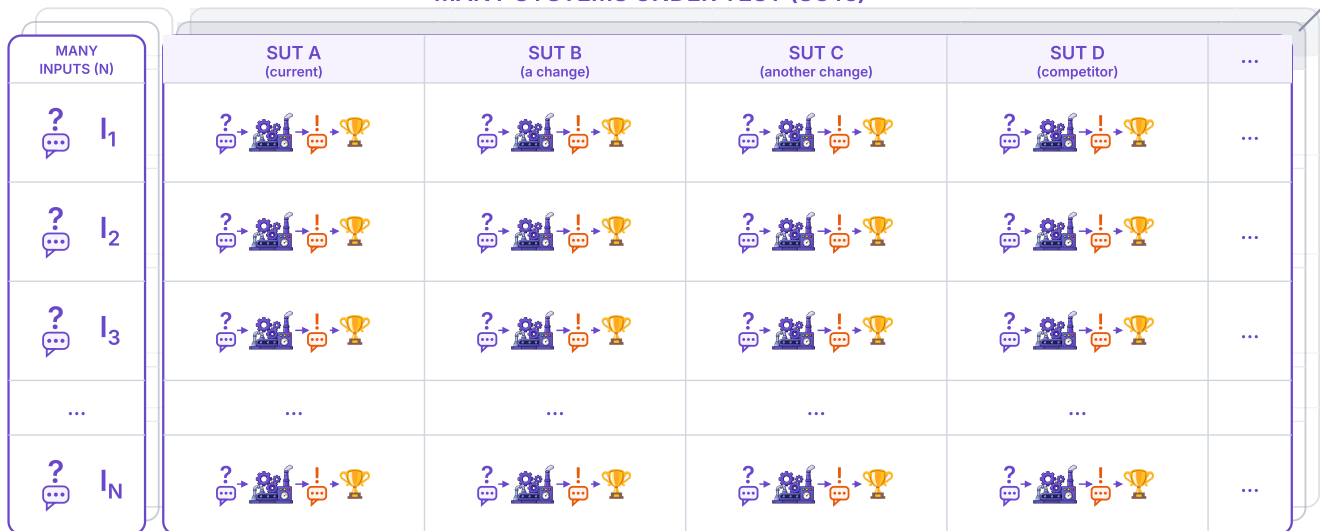
Where we are

- ✓ Section 1 — What we do today — and the problem with it
- ▶ **Section 2 — What the noise looks like, and how we measure it**
- Section 3 — Should we deploy?
- Section 4 — How much data do we need?

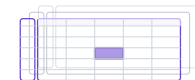
Some terminology



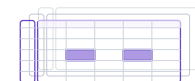
MANY SYSTEMS UNDER TEST (SUTs)



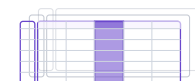
NAMING THE SLICES



example
one metric



paired example
one input, two SUTs



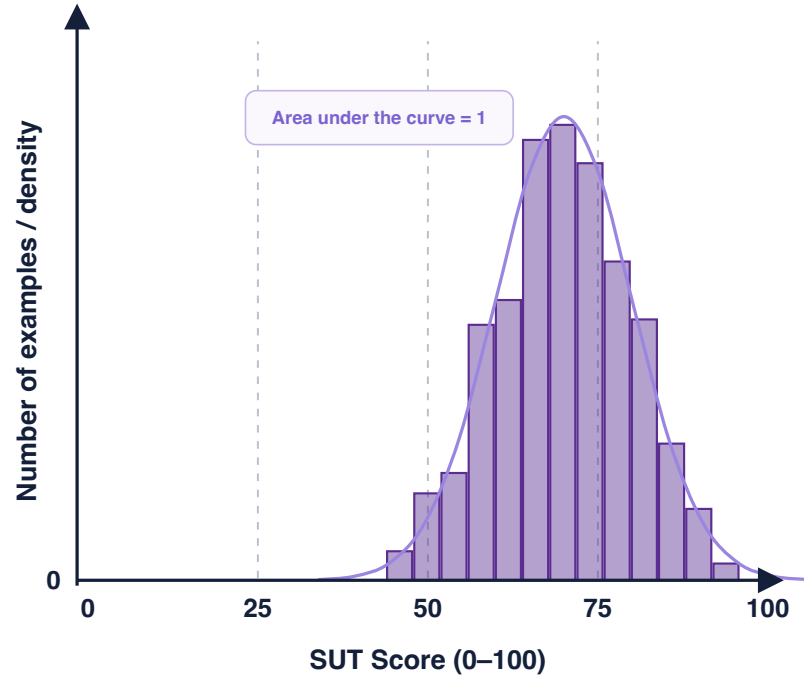
experiment / sample
N inputs, one SUT



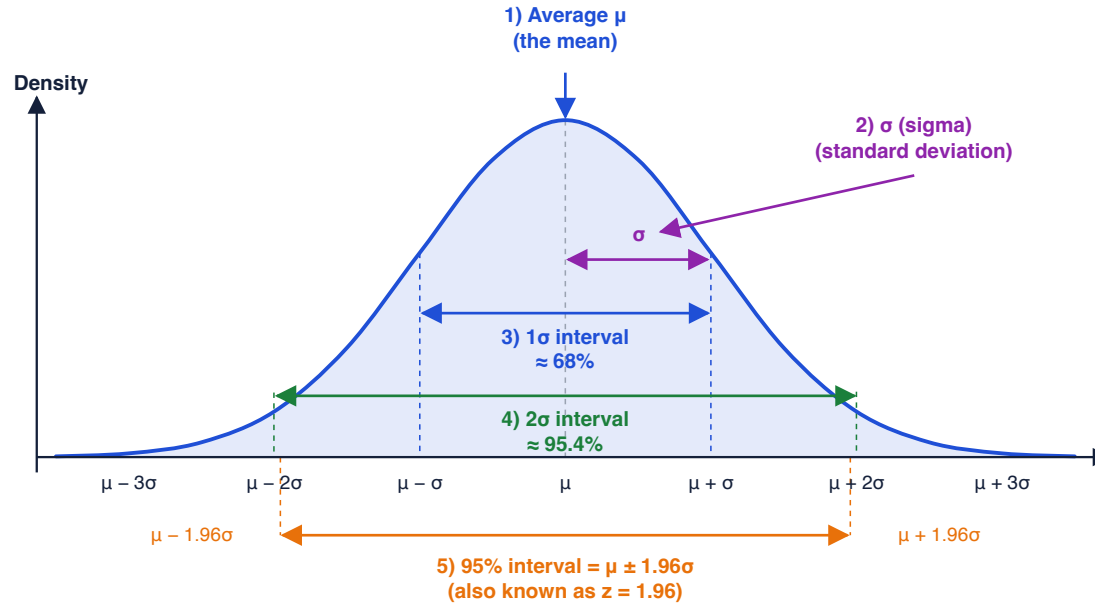
replication
same experiment, every re-run

From scores to a distribution

Example ID (i)	SUT Score (0–100)
1	82.1
2	67.3
3	77.8
4	91.6
5	61.4
...	...
$N - 2$	45.2
$N - 1$	72.9
N	88.7
mean	70.0



Anatomy of a distribution



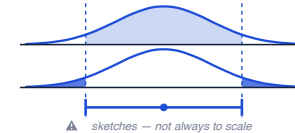
Mean (μ)

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

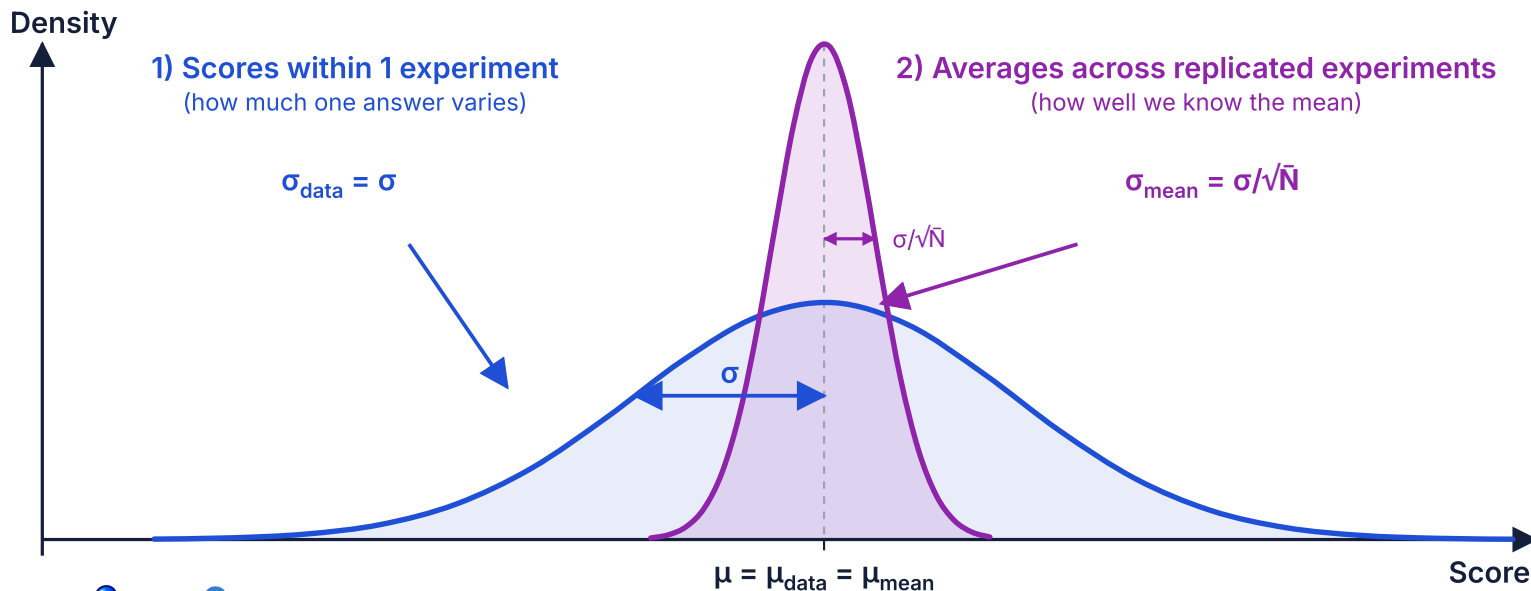
Standard Deviation (σ)

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

3 ways to depict the 95% interval



Two spreads: the data vs the average



But which one do we care about?

We want to deploy based on a reproducible decision,
so we care about variance across replications.

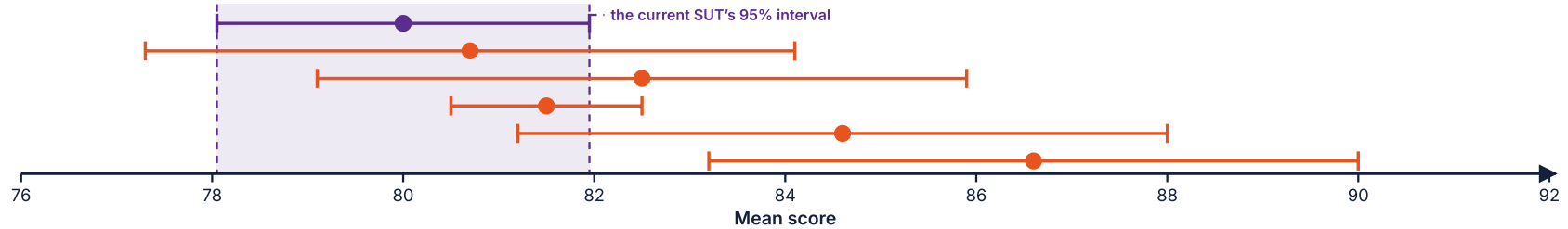
So we need σ_{mean} !



Where we are

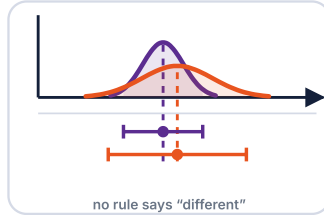
- ✓ Section 1 — What we do today — and the problem with it
- ✓ Section 2 — What the noise looks like, and how we measure it
- ▶ **Section 3 — Should we deploy?**
- Section 4 — How much data do we need?

Add a statistical perspective: every score is an estimate — let's add error bars.

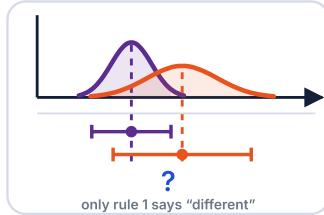


there are 3 rules one thinks of quite quickly

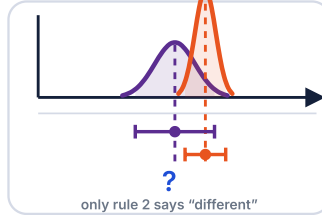
each mean inside the other's



RULE 1
new mean outside current's



RULE 2
current mean outside new's



RULE 3
means outside, overlapping



intervals don't overlap



Not significant

Significant or not? Depends which rule you pick.

Significantly better

The actual rule



We need σ_Δ , the spread of the difference.

It depends on how the two SUTs correlate: ρ .

$$\sigma_\Delta = \sqrt{\sigma_A^2 + \sigma_B^2 - 2\rho\sigma_A\sigma_B}$$

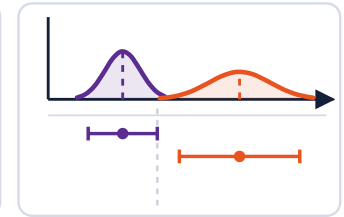
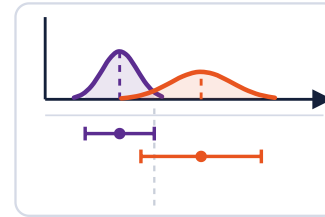
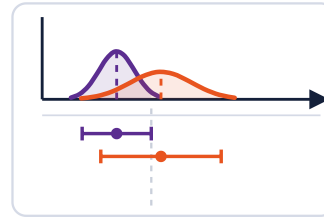
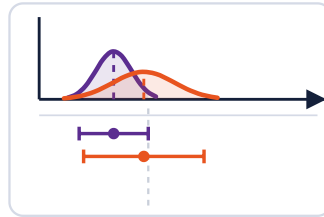
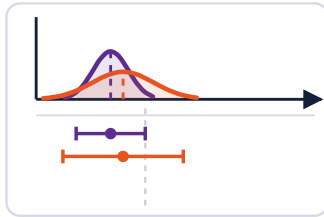
95% interval: $\mu_\Delta \pm 1.96 \sigma_\Delta / \sqrt{N}$

THE FLOOR

perfect correlation

$$\rho = +1$$

$$\sigma_\Delta = |\sigma_A - \sigma_B|$$



definitely not significant

depends on ρ

clearly significant

RULES 1 & 2

one rule, not two
you are probably near here

$$\rho = +0.87$$

$$\sigma_\Delta \rightarrow \min(\sigma_A, \sigma_B)$$

THE SIMPLEST CASE

no correlation

$$\rho = 0$$

$$\sigma_\Delta = \sqrt{\sigma_A^2 + \sigma_B^2}$$

RULE 3

no overlap at all
perfect inverse correlation

$$\rho = -1$$

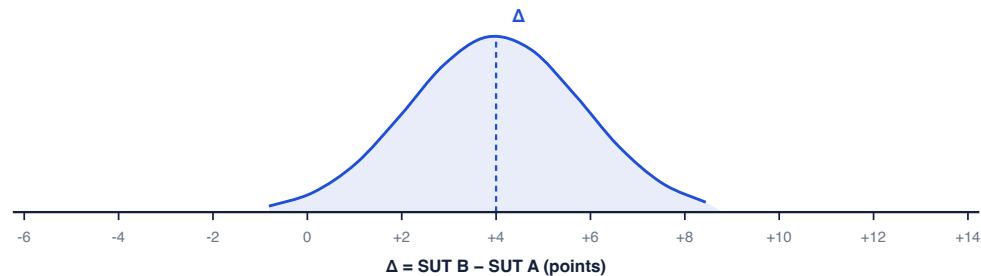
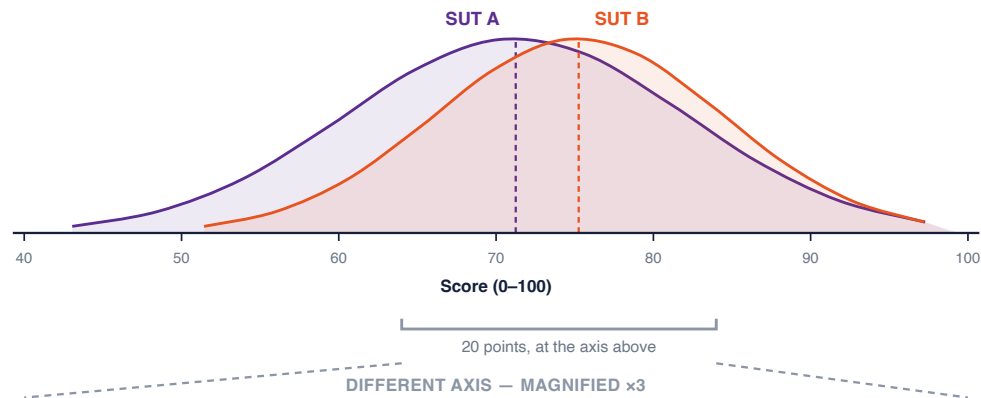
$$\sigma_\Delta = \sigma_A + \sigma_B$$

Run both systems on the same examples, then subtract

Example ID (i)	SUT A	SUT B	$\Delta = B - A$
1	62	67	+5
2	81	84	+3
3	74	77	+3
4	55	62	+7
5	88	90	+2
...	...	...	...
N - 2	69	73	+4
N - 1	77	79	+2
N	64	70	+6
mean $\pm \sigma$	71.2 $\pm$ 10.8	75.2 $\pm$ 9.2	4.0 $\pm$ 1.9



Pairing saves us having to figure out ρ

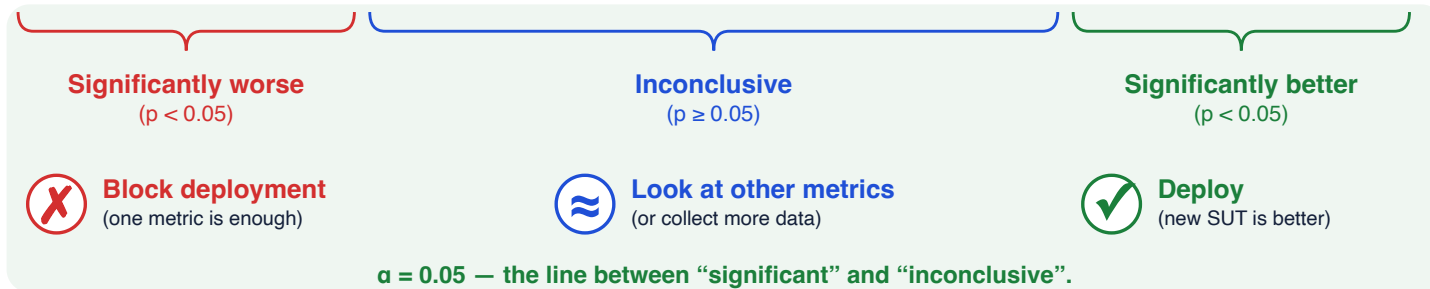


The balanced rule

Common rule



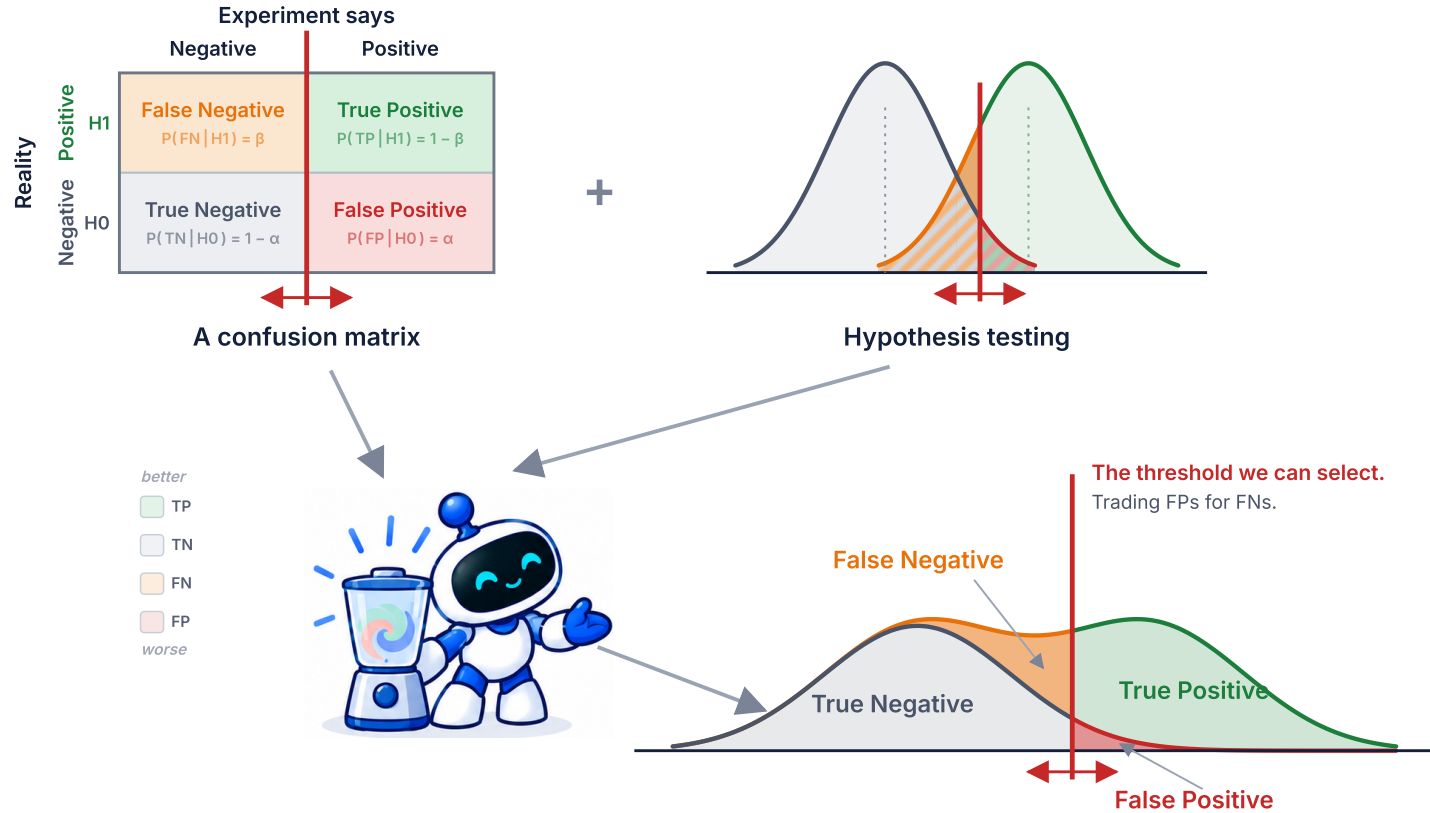
Balanced rule



Where we are

- ✓ Section 1 — What we do today — and the problem with it
- ✓ Section 2 — What the noise looks like, and how we measure it
- ✓ Section 3 — Should we deploy?
- ▶ **Section 4 — How much data do we need?**

The continuous confusion matrix

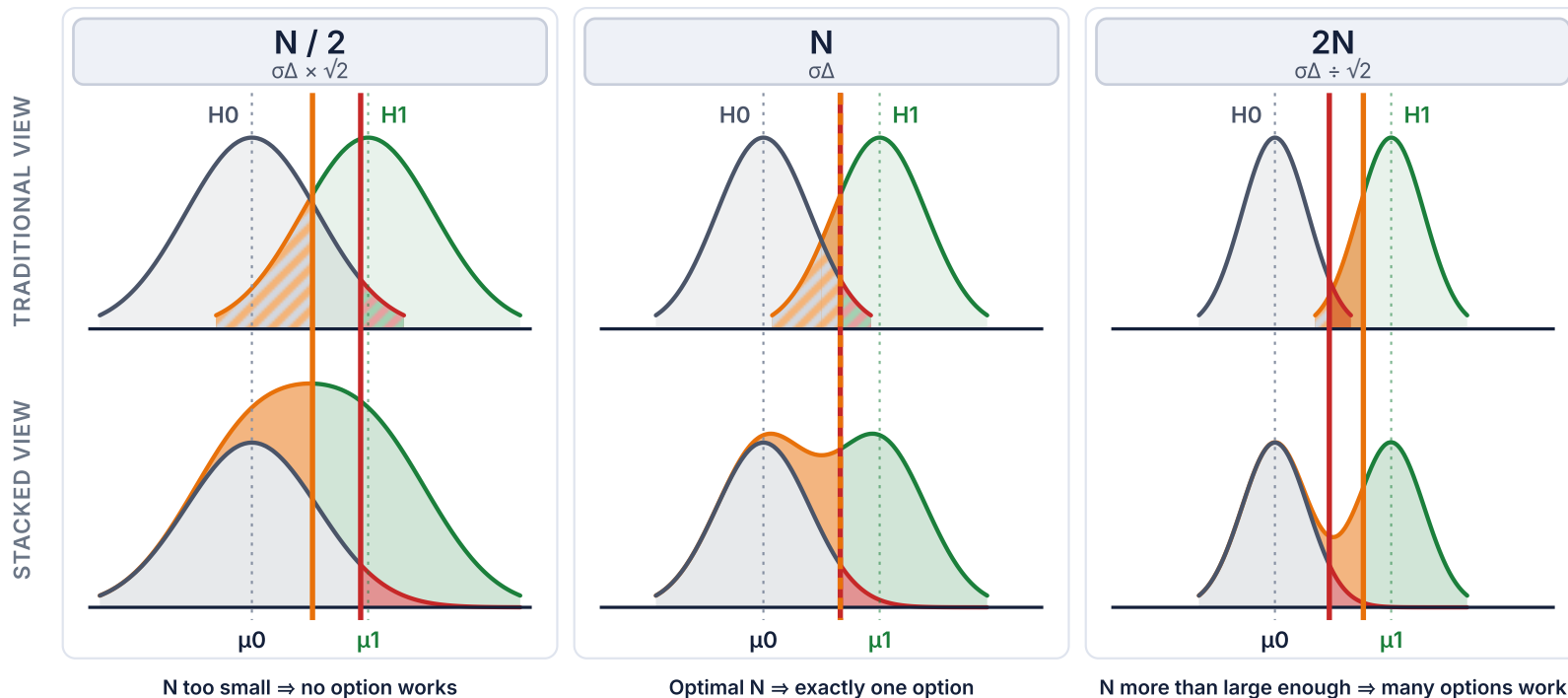


We can't control α and β independently, right?

Or maybe we can 🤔 we just need a new degree of freedom to vary.

— α threshold (5%)

— β threshold (20%)



The N→reject formula

YOU CHOOSE THESE

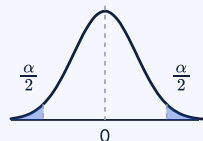
1 $z_{\alpha/2} = 1.96$

How often will I tolerate a regression in prod?

$\alpha = 5\%$, two-sided

2.5% deploy a non-improvement

2.5% block a fine SUT

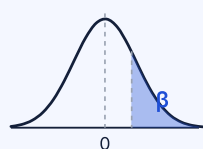


	Fail	Reject
H1: diff	False Negative	True Positive
H0: none	True Negative	False Positive

2 $z_{1-\beta} = 0.84$

How often will I tolerate a missed opportunity?

$\beta = 20\%$, one-sided → power = 80%



	Fail	Reject
H1: diff	False Negative	True Positive
H0: none	True Negative	False Positive

$$N = \left(\frac{(z_{\alpha/2} + z_{1-\beta}) \sigma_{\Delta}}{|\Delta|} \right)^2$$

N = number of paired questions needed



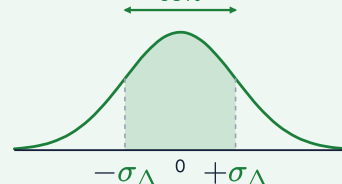
YOU MEASURE THIS

3 σ_{Δ}

How noisy is the comparison?

measured from your pilot sample

68%



YOUR CALL

4 $|\Delta|$

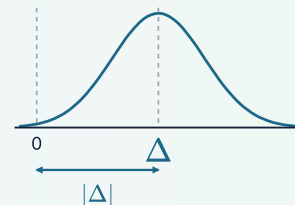
How big is the difference?

choose it

smallest worth detecting

measure it

observed in your pilot sample



Example results

Current SUT mean total: 90.00 ± 2.80 (mean $\pm$ half-width 95% CI, N = 200, scale 0–100)

Scenario (new SUT)	Mean total (0–100) [95% CI]	N (dataset)	$\mu\Delta \pm 1.96 \cdot \sigma\Delta / \sqrt{N}$ [95% CI]	p-value	Significant (p < 0.05)?	Needed power (N reject)
Clear-cut						
significantly better <i>deploy the new SUT</i>	95.00 ± 2.45 [92.55, 97.45]	200	$+5.00 \pm 2.08$ [+2.92, +7.08]	< 0.0001	✓	71
significantly worse <i>block the deployment</i>	85.50 ± 3.05 [82.45, 88.55]	200	-4.50 ± 2.56 [-7.06, -1.94]	0.0006	✓	133
Not clear-cut						
seems better, but needs more data <i>look at other metrics, or collect more data</i>	92.50 ± 2.90 [89.60, 95.40]	200	$+2.50 \pm 2.66$ [-0.16, +5.16]	0.0656	✗	463
seems worse, but needs more data <i>look at other metrics, or collect more data</i>	88.00 ± 2.75 [85.25, 90.75]	200	-2.00 ± 2.77 [-4.77, +0.77]	0.1573	✗	785
deeply non-significant <i>call them equivalent; decide on cost or latency</i>	89.50 ± 2.60 [86.90, 92.10]	200	-0.50 ± 2.43 [-2.93, +1.93]	0.6862	✗	9615
Clear-cut, but tricky						
significant, but underpowered <i>deploy — but it may not replicate</i> <i>re-run it and the verdict flips: power is only 63%</i>	<i>run 1</i> 93.50 ± 3.20 [90.30, 96.70]	200	$+3.50 \pm 3.01$ [+0.49, +6.51]	0.0225	✓	302
	<i>re-run</i> 92.10 ± 3.10 [89.00, 95.20]	200	$+2.10 \pm 2.85$ [-0.75, +4.95]	0.1494	✗	756

Δ = new SUT mean – current SUT mean (positive = better).

Needed power = N for 80% power at $\alpha = 0.05$ to detect the observed Δ .

NUMBERS FAKE
FOR ILLUSTRATIVE PURPOSES

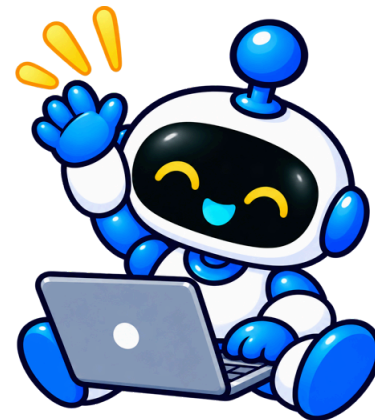
Where we are

- ✓ Section 1 — What we do today — and the problem with it
- ✓ Section 2 — What the noise looks like, and how we measure it
- ✓ Section 3 — Should we deploy?
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Get the slides here!

Questions? — Thank you



Is it fair to assume normal distributions?

Central Limit Theorem: **distribution of cases** → **distribution of averages**



= a single experiment (the cases)

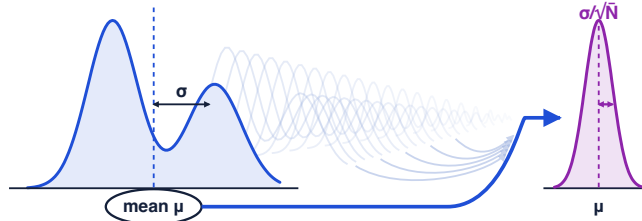


= repeated experiments

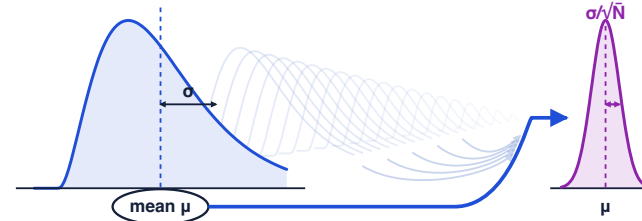


= the distribution of their averages

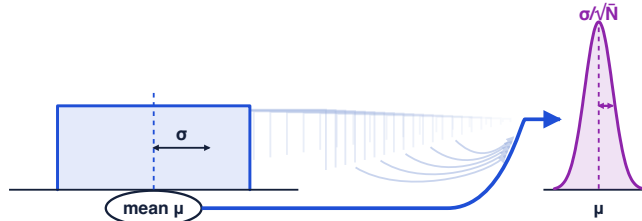
1) Multimodal (bimodal)



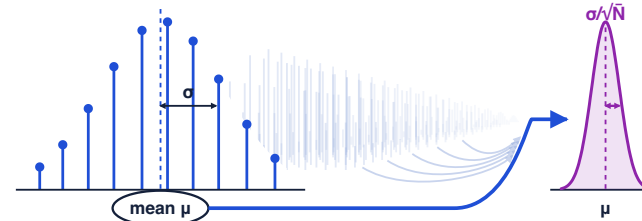
2) Skewed (right-skewed)



3) Uniform (all scores equally likely)



4) Discrete (0–1 scores)



What this means

- Your weird score shapes are fine — the distribution of the **average** is still $\approx$ Normal.
- Many re-runs (M) are how you map that distribution out — each run is one draw from it.
- Its width is $\sigma/\sqrt{N}$: more questions narrow it; the scores' own spread σ doesn't change.



Not shown: heavy tails — usually fine too, just hard to draw at this scale. (Some infinite-variance ones don't satisfy the CLT.)

Every rule is a hidden ρ

$$\sigma_{\Delta} = \sqrt{\sigma_A^2 + \sigma_B^2 - 2\rho\sigma_A\sigma_B}$$

ρ = how much the two systems' scores move together, question by question.

Pick a rule and you have picked a ρ — whether you meant to or not.

illustrative numbers

$\rho = +1$
PERFECTLY IN STEP

$$\sigma_{\Delta} = |\sigma_A - \sigma_B|$$

$$\Delta = 1.45$$

Every question that is hard for one system is hard for the other. The shared difficulty cancels — this is the best pairing can ever do.

$\rho = 0$
INDEPENDENT

$$\sigma_{\Delta} = \sqrt{(\sigma_A^2 + \sigma_B^2)}$$

$$\Delta = 3.92$$

No shared difficulty at all.
This is what the actual rule assumes.

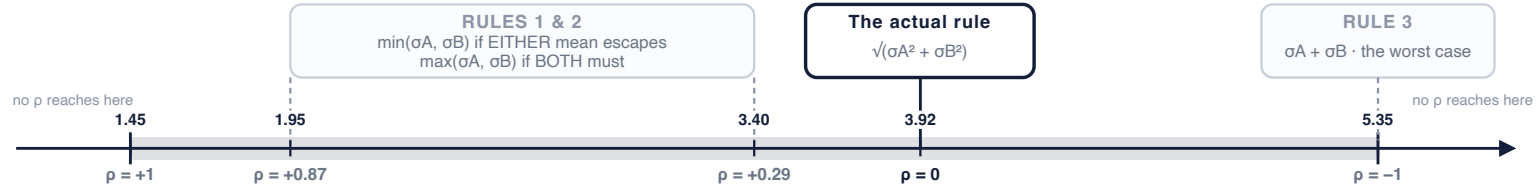
$\rho = -1$
PERFECTLY OPPOSED

$$\sigma_{\Delta} = \sigma_A + \sigma_B$$

$$\Delta = 5.35$$

One system's easy questions are the other's hard ones. The worst case — and exactly rule 3.

σ_{Δ} can only land in this band — and each rule sits at one point in it



In practice $\rho > 0$ — the same hard questions hurt both systems — so the truth sits left of $\rho = 0$. Pairing the answers measures it instead of assuming it.

Rules 1 & 2 treat σ_{Δ} as max(σ_A , σ_B) · Rule 3 treats it as $\sigma_A + \sigma_B$ · The actual rule treats it as $\sqrt{(\sigma_A^2 + \sigma_B^2)}$

Three rules, one formula, three different assumptions about ρ .

ROC curve: more data moves you from guessing to knowing

